Acceptance test: BLASTP on default Uniprot database
As a life science scientist

I want to perform BLASTP using the default Uniprot database after I entered either a protein sequence in FASTA format in the therefore intended text box or uploaded a FASTA file of a protein sequence. After entering the sequence, I want to press the BLAST button to run the BLASTP.|

So that I can analyze, using the BLASTP output, which proteins are similar to my query sequence and from what organism these proteins are.|

Acceptance criteria:

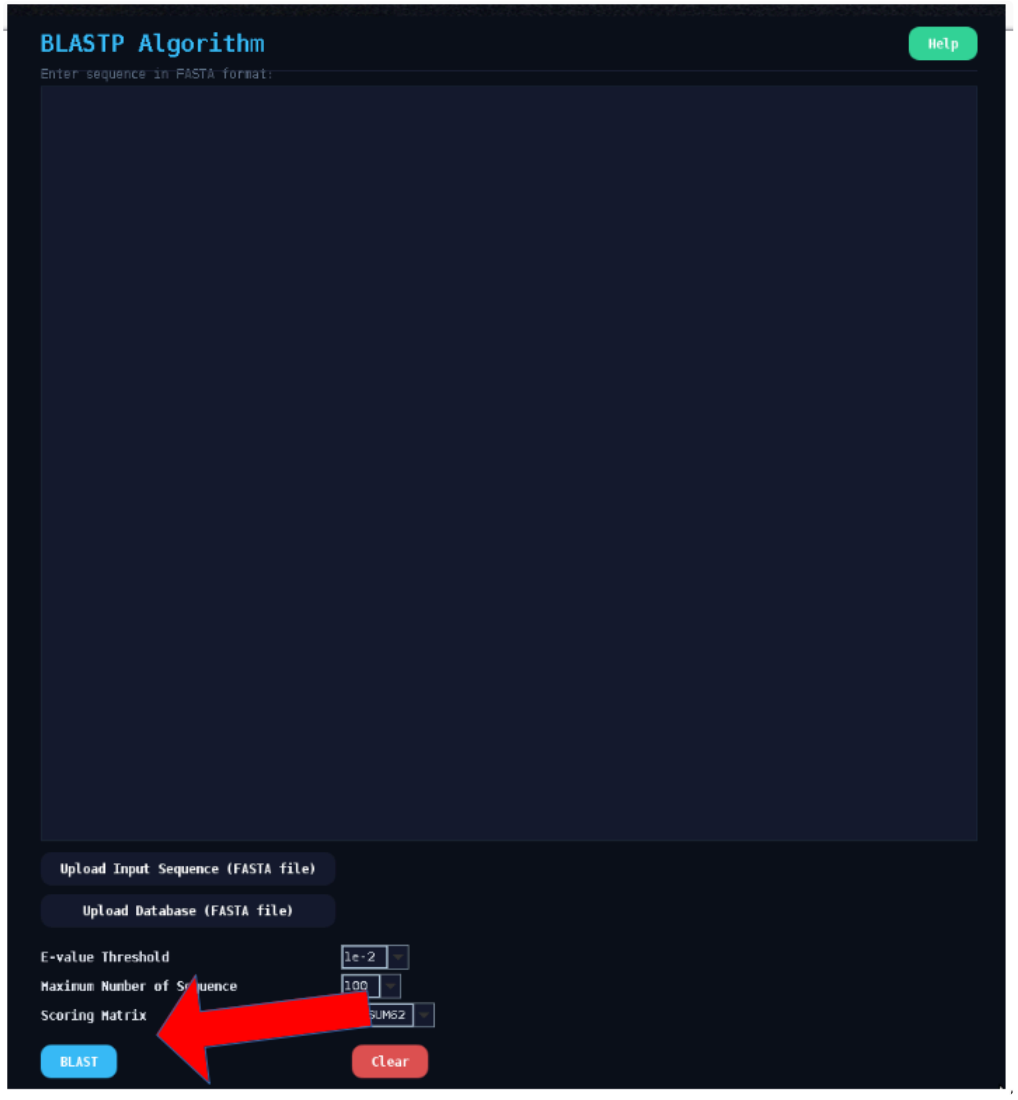
- After I put in either a protein sequence in FASTA format in the therefore intended text box or upload a FASTA file of a protein sequence and press the BLAST button, the BLASTP output should be written to a file in my downloads folder and shown in a new window.
- Error message saying that the user has to either enter a protein in FASTA file format in the text box or upload a query FASTA file when the user presses the BLAST button without providing a query sequence.

For this acceptance test the DNA and protein sequence of the protein 'Green fluorescent protein' is used (a protein commonly used as fluorescent tag in life science experiments)

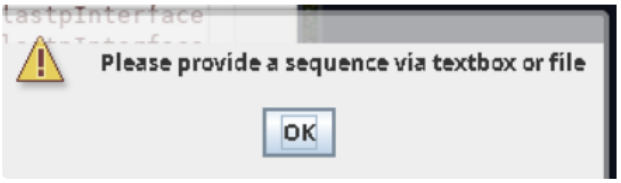
```
>GFP
MSKGEELFTGVVPILVELDGDVNGHKFSVSGEGEGDATYGKLTLKFICTTGKLPVPWPTLVTTFSYGVQCFSRYPDHMKQHDFFKSAMPE
GYVQERTIFFKDDGNYKTRAEVKFEGDTLVNRIELKGIDFKEDGNILGHKLEYNYNSHNVYIMADKQKNGIKVNFKIRHNIEDGSVQLAD
HYQQNTPIGDGPVLLPDNHYLSTQSALSKDPNEKRDHMVLLEFVTAAGITHGMDELYK
```

Click on the BlastP when the program opens

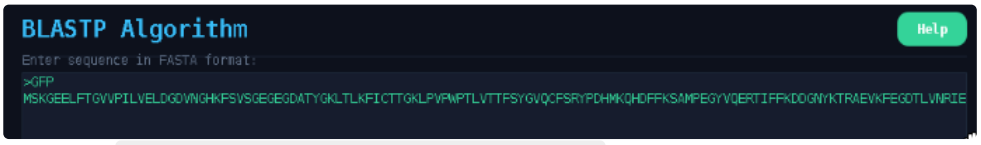
The blast GUI is displayed with a textbox for the file



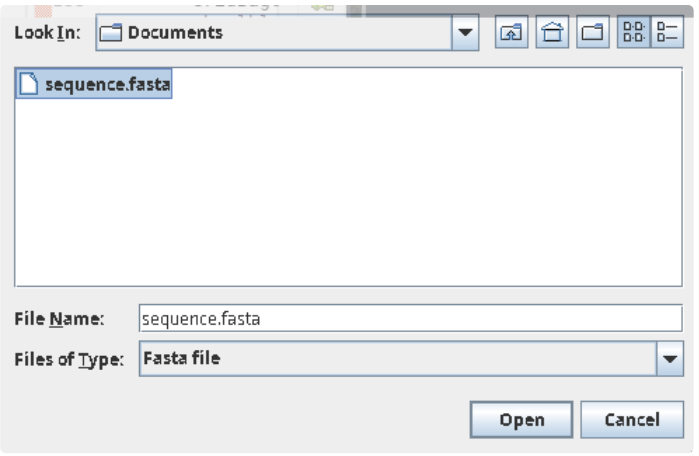
When the blast button is clicked without giving a sequence in the text box and error message appears



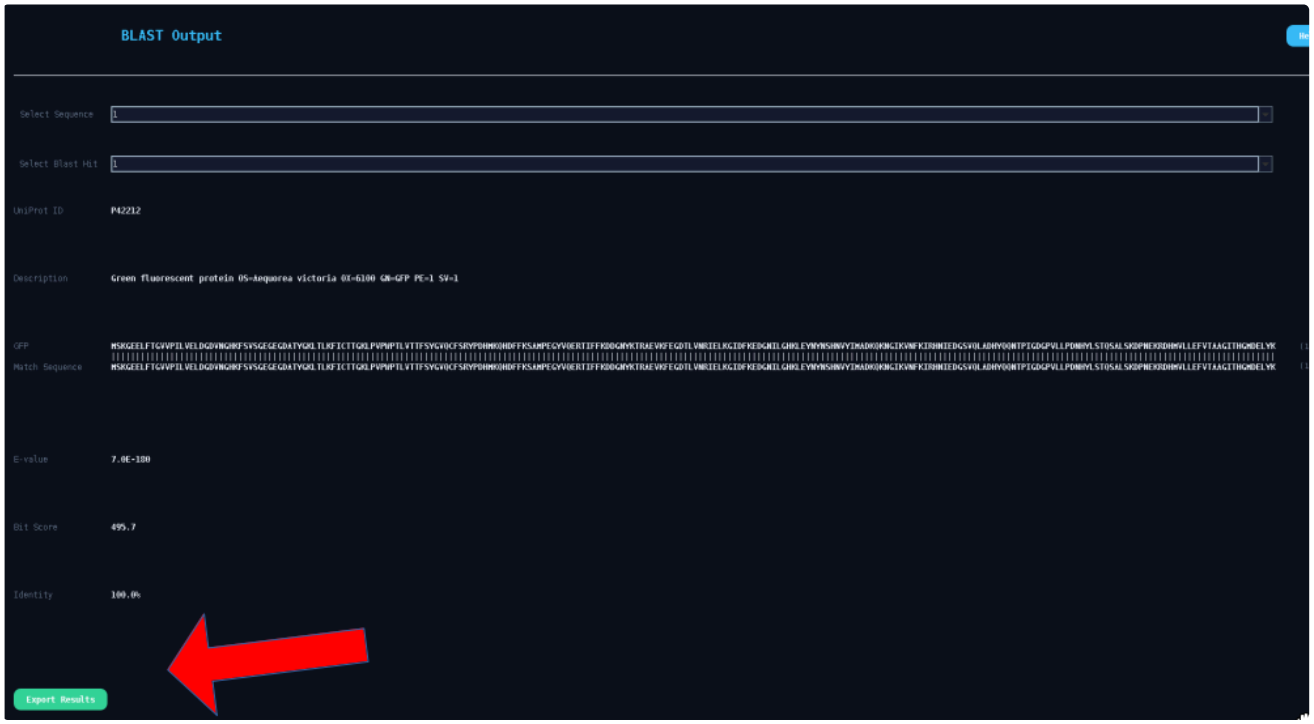
GFP sequence can be inserted the text box



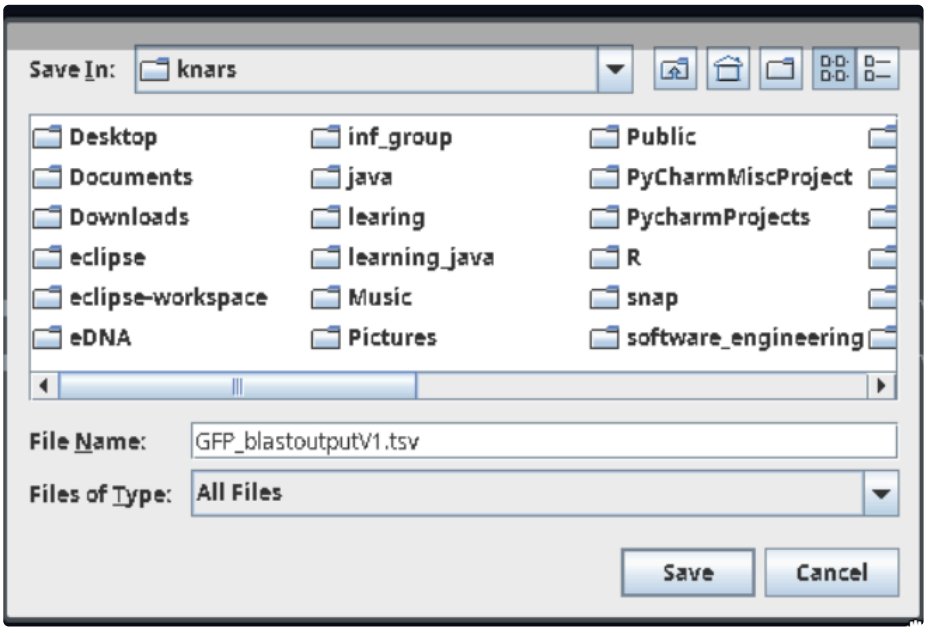
Or the `upload Input Sequence (FASTA file)` button is clicked to upload a fasta file



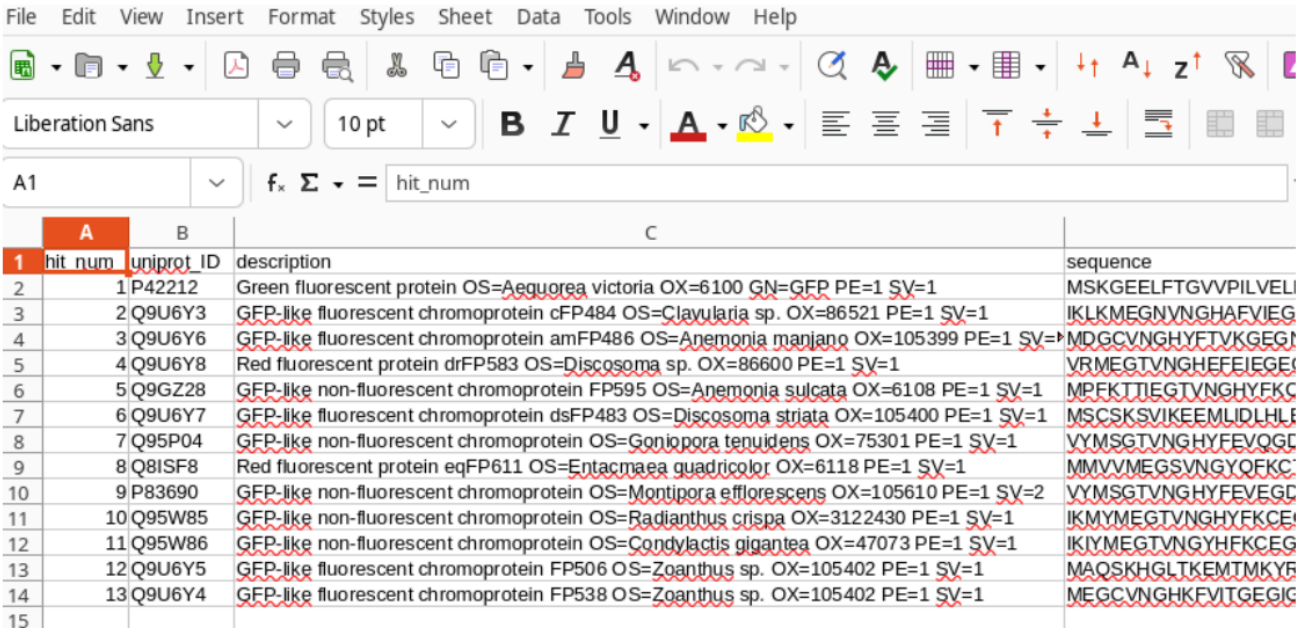
When then the blast button is clicked again the results are correctly displayed in a results GUI and can be exported to a TSV file by clicking on the export results button



Location of the file can be chosen



TSV file contains the corrects information



	A	B	C	
1	hit_num	uniprot_ID	description	sequence
2	1	P42212	Green fluorescent protein OS=Aequorea victoria OX=6100 GN=GFP PE=1 SV=1	MSKGEELFTGVVPILVEL
3	2	Q9U6Y3	GFP-like fluorescent chromoprotein cFP484 OS=Clavularia sp. OX=86521 PE=1 SV=1	IKLKMEGNVNGHAFVIEG
4	3	Q9U6Y6	GFP-like fluorescent chromoprotein amFP486 OS=Anemonia manjano OX=105399 PE=1 SV=	MDGCVNGHYFTVKGEGN
5	4	Q9U6Y8	Red fluorescent protein drFP583 OS=Discosoma sp. OX=86600 PE=1 SV=1	VRMEGTVNGHEFEIEGE
6	5	Q9GZ28	GFP-like non-fluorescent chromoprotein FP595 OS=Anemonia sulcata OX=6108 PE=1 SV=1	MPFKTTIEGTVNGHYFKC
7	6	Q9U6Y7	GFP-like fluorescent chromoprotein dsFP483 OS=Discosoma striata OX=105400 PE=1 SV=1	MSCSKSVIKEEMLIDLHL
8	7	Q95P04	GFP-like non-fluorescent chromoprotein OS=Goniopora tenuidens OX=75301 PE=1 SV=1	VYMSGTVNGHYFEVQGD
9	8	Q8ISF8	Red fluorescent protein eqFP611 OS=Entacmaea quadricolor OX=6118 PE=1 SV=1	MMVVMEGSVNGYQFKC
10	9	P83690	GFP-like non-fluorescent chromoprotein OS=Montipora efflorescens OX=105610 PE=1 SV=2	VYMSGTVNGHYFEVEGD
11	10	Q95W85	GFP-like non-fluorescent chromoprotein OS=Radianthus crispa OX=3122430 PE=1 SV=1	IKMYMEGTVNGHYFKCE
12	11	Q95W86	GFP-like non-fluorescent chromoprotein OS=Condylactis gigantea OX=47073 PE=1 SV=1	IKIYMEGTVNGYHFKCEG
13	12	Q9U6Y5	GFP-like fluorescent chromoprotein FP506 OS=Zoanthus sp. OX=105402 PE=1 SV=1	MAQSKHGLTKEMTMKYR
14	13	Q9U6Y4	GFP-like fluorescent chromoprotein FP538 OS=Zoanthus sp. OX=105402 PE=1 SV=1	MEGCVNGHKFVITGEGI
15				